
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is a central requirement for deploying machine
2 learning systems in settings where test inputs may extend beyond the training
3 distribution. Unlabeled data collected in deployment or other open environments
4 offers a broadly useful signal for this problem, but its value is difficult to realize
5 because such data typically mixes in-distribution (InD) and OOD samples rather
6 than providing a clean auxiliary OOD set. This mixed-data setting makes it
7 challenging to train an OOD classifier without imposing strong assumptions about
8 which unlabeled examples are truly outlying. In this work, we introduce Medix,
9 a median-based framework for extracting candidate outliers from unlabeled wild
10 data. The median provides a robust estimate of central tendency in gradient space,
11 making it a natural mechanism for identifying samples that shift the wild-data
12 gradient profile away from the InD reference. Medix then combines the extracted
13 candidates with labeled InD data to train an OOD detector. Theoretically, we derive
14 error bounds showing that the filtering stage can maintain low error under the
15 stated contamination and concentration conditions. Empirically, Medix improves
16 over existing methods across the evaluated open-world benchmarks, supporting the
17 broader use of median-centric filtering for OOD detection with unlabeled mixtures.

18 1 Introduction

19 Machine learning models deployed beyond curated training environments commonly encounter inputs
20 that differ from their training distribution, creating safety and reliability challenges across application
21 domains. Such out-of-distribution (OOD) samples may correspond to categories or conditions absent
22 from the label space, and the model should avoid treating them as ordinary in-distribution cases.
23 Standard neural networks, however, often lack an explicit mechanism for recognizing and handling
24 these inputs at inference time (Nguyen et al., 2015).

25 Detecting OOD inputs at inference time is difficult precisely because the relevant unknown distri-
26 butions are not observed as labeled classes during training. A growing line of work (Katz-Samuels
27 et al., 2022a; Du et al., 2024a) therefore studies whether additional unlabeled “in-the-wild” data can
28 help bridge this gap. In particular, Katz-Samuels et al. (2022a) use unlabeled wild data to regularize
29 training while preserving the objective of classifying labeled InD samples. The appeal of this setting
30 is that wild data can often be collected after deployment, from the same broad operating environment
31 in which future OOD inputs may arise. When used appropriately, such data can expose the detector
32 to a wider range of deployment-relevant variation than the labeled InD set alone.

33 The central difficulty is that wild data is not a labeled OOD resource; it is an unlabeled mixture whose
34 samples may come from either the InD distribution or an OOD source. This ambiguity limits the
35 direct applicability of methods that assume clean auxiliary outliers and calls for filtering mechanisms
36 that can operate at the dataset level without sample labels. Despite its practical relevance, this mixed
37 wild-data formulation remains comparatively underexplored. Formal guarantees are also scarce,

38 and to the best of our knowledge, Du et al. (2024a) is the only prior work that gives a theoretical
 39 foundation for the “in-the-wild” OOD setting.

40 At the same time, median-based procedures have shown promise as broadly robust tools for data
 41 pruning (Acharya et al., 2024). Motivated by this connection between robust statistics and sample
 42 selection, we study the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD
 detection with theoretical guarantees?

43 We answer this question by developing a median-centric view of OOD detection from unlabeled
 44 mixtures. Concretely, we formulate a median-based optimization problem and instantiate it through
 45 Medix, a greedy filtering algorithm that identifies candidate OOD samples from wild data with low
 46 error under the assumptions we analyze. Our theory supports this broader framing by bounding the
 47 fraction of InD samples mistakenly flagged as outliers in the unlabeled “in-the-wild” setting. The
 48 bound separates two general effects: a contamination effect, which captures how OOD points influence
 49 the median and remains controlled when the OOD proportion is below 50%; and a concentration
 50 effect, which uses sub-Gaussian InD gradients to control deviations from the InD mean gradient.
 51 Together, these terms show that median-based filtering can remain reliable for mixed unlabeled
 52 data under the stated conditions, and our experiments further examine this behavior across multiple
 53 benchmark configurations.

54 We evaluate Medix against two broad families of OOD detectors: methods trained only on InD data
 55 and methods that also use an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky et al., 2009),
 56 Medix improves over the strong KNN+ baseline (Sun et al., 2022) by an average of 40.98% in FPR95.
 57 Unlike Outlier Exposure (Hendrycks et al., 2019) and related methods that require a clean auxiliary
 58 set separable from the InD data, Medix is designed for the weaker and more broadly applicable case
 59 where auxiliary data is mixed. Relative to WOODS (Katz-Samuels et al., 2022a), Medix reduces the
 60 average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10.

61 Our key contributions are as follows:

- 62 **C1)** We propose Medix, a median-centric greedy approach that filters outliers from the unlabeled
 63 wild data and then trains an OOD detector on the identified outliers and the InD samples.
 64 Our main contribution is the filtering stage.
- 65 **C2)** We establish theoretical guarantees for the robustness of median-based filtering in identi-
 66 fying both inliers and outliers within unlabeled mixtures. Specifically, we prove that the
 67 misclassification rates for both InD samples flagged as outliers and OOD samples retained
 68 as inliers are tightly controlled by two effects: the contamination effect, which remains
 69 bounded as long as the proportion of OOD samples is below 50%, and the concentration
 70 effect, which ensures stability in the gradient behavior of InD samples.
- 71 **C3)** We evaluate Medix across eleven InD-OOD pairs and compare it with 20 competitive
 72 baselines spanning InD-only and wild-data training regimes. Across these evaluated settings,
 73 Medix achieves the best overall performance among the compared methods. We also
 74 empirically examine the filtering stage itself and find that Medix can extract outliers with a
 75 low error rate (e.g. only 12.5%; see Figure 2), consistent with the theoretical picture.

76 2 Preliminaries and Problem Setup

77 This section formalizes the general wild-data OOD detection setting considered throughout the paper,
 78 including the data distributions, model classes, loss functions, and evaluation objectives.

79 **Labeled In-Distribution Data.** Let $\mathcal{X}$ denote the input space and $\mathcal{Y} = \{1, \dots, K\}$ the label space for
 80 the InD task, so that the same notation can cover any K -class supervised setting. The labeled dataset
 81 $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ consists of n i.i.d. samples from an unknown joint distribution $\mathbb{P}_{XY}$
 82 over $\mathcal{X} \times \mathcal{Y}$. Its marginal on $\mathcal{X}$ is denoted by $\mathbb{P}_{\text{in}}$ and represents the InD input distribution for the
 83 task. We train the base InD model using $\mathcal{S}_{\text{in}}$.

84 **Out-of-distribution detection.** We consider the deployment regime in which a model trained on
 85 labeled InD data may later receive inputs associated with classes or sources outside the training label
 86 set, i.e., for some label $y \notin \mathcal{Y}$. For such inputs, the desired behavior is not ordinary classification but

abstention or rejection. Thus, the inference-time goal is to decide whether an input is compatible with $\mathbb{P}_{\text{in}}$ or should be treated as arising from an OOD source.

Unlabeled wild data. A major barrier to OOD detection is that labeled OOD examples are rarely available at the scale or diversity needed to represent possible deployment shifts. We therefore incorporate unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, following the practical formulation of Katz-Samuels et al. (2022a). Wild data may be gathered from the operating environment of a model trained on $\mathcal{S}_{\text{in}}$, and it is treated as a mixture rather than as a clean OOD set. Following Du et al. (2024a); Katz-Samuels et al. (2022a), we model this mixture through the Huber contamination model (Huber, 1964)

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π is the contamination proportion and $\mathbb{P}_{\text{out}}$ denotes the OOD input distribution over $\mathcal{X}$. The case $\pi = 0$ contains no OOD samples and therefore reduces to a trivial instance of the filtering problem.

Models and Loss Functions. Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ be the InD classifier with parameters $\phi \in \Phi$, where Φ is the corresponding parameter space. Its outputs define scores over the $K = |\mathcal{Y}|$ InD classes, and the labeled InD loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we use a separate binary detector $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ with parameters $\theta \in \Theta$. The associated binary loss is denoted by $\ell_b(g_\theta(x), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, with $y_+ > 0$ representing InD inputs and $y_- < 0$ representing OOD inputs.

Learning objective. The learning objective uses both labeled InD data $\mathbb{P}_{\text{in}}$ and unlabeled wild data $\mathbb{P}_{\text{wild}}$ to support a detector g_θ while preserving the multi-class classifier f_ϕ . At test time, performance is evaluated with the following standard quantities:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X}\mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ denotes the OOD distribution used for test evaluation, which need not be identical to the wild OOD component in later experiments.

3 Method: Median-Centric Framework for OOD Detection

This section presents **Medix**, a learning framework for OOD detection that uses unlabeled wild data without requiring it to be a clean outlier set. The framework is intended for open-world deployment regimes in which models may encounter previously unseen inputs and the available auxiliary data may be mixed. Medix has two stages: **1) Outlier Extraction**: a filtering procedure that selects candidate OOD samples from the unlabeled wild data (Section 3.1), and **2) Detector Training**: training a binary OOD detector from labeled InD data together with the extracted candidates (Section 3.2). For the second stage, we follow the protocol of Du et al. (2024a). This decomposition separates the broadly useful task of filtering unlabeled mixtures from the downstream task of detector learning, allowing the method to leverage wild data while keeping the experimental setting and assumptions explicit.

3.1 Extracting Candidate Outliers from the Wild Data

To extract candidate outliers from the mixed set $\mathcal{S}_{\text{wild}}$, Medix uses an optimization-based criterion in the gradient space of the model parameters. The gradients are computed from a classifier f_ϕ trained only on the labeled InD set $\mathcal{S}_{\text{in}}$, so the filtering signal is anchored to information available before observing any OOD labels. We formalize this procedure below.

Reference gradient estimation from InD data. The first step estimates an InD reference gradient from $\mathcal{S}_{\text{in}}$. We train the classifier f_ϕ on the labeled InD data using empirical risk minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training, we summarize the InD gradient profile by averaging loss gradients with respect to the model parameters over the labeled InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

128 In Medix, $\bar{\nabla}_{\text{in}}$ is the reference point against which gradients from wild samples are compared.

129 **Motivation.** The guiding hypothesis is that adding OOD samples to a wild mixture shifts the
 130 robust aggregate of wild gradients away from the InD reference $\bar{\nabla}_{\text{in}}$. We examine this behavior in
 131 a preliminary CIFAR-10/SVHN study, using CIFAR-10 (Krizhevsky et al., 2009) as InD data and
 132 SVHN (Netzer et al., 2011) as OOD data. The initial $\mathcal{S}_{\text{wild}}$ contains 10,000 CIFAR-10 samples
 133 disjoint from the training set used to learn $\phi_{\mathcal{S}_{\text{in}}}$ and compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN
 134 samples and measure the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of
 135 wild-sample gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

136 Figure 1 shows a clear monotonic increase in this deviation as more OOD samples are included,
 137 supporting the use of median-gradient deviation as a filtering signal in this benchmark setting. This
 138 trend also motivates the stopping rule used by Medix: the iterative procedure halts when the change
 139 in L_2 deviation between consecutive iterations falls below a threshold ϵ .

140 **Filtering potential outliers from unlabeled wild data.** Motivated by the deviation pattern in Figure
 141 1, we formulate a dataset-level optimization problem that seeks the InD-like subset of the wild mixture
 142 and thereby identifies the complementary outlier subset $\mathcal{S}_{\text{out}}^*$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

143 Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted
 144 label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$.
 145 This objective searches for a subset $\mathcal{S}$ of $\mathcal{S}_{\text{wild}}$
 146 whose EWM gradient is closest to the InD reference
 147 $\bar{\nabla}_{\text{in}}$. Under the mixture interpretation sup-
 148 ported by Figure 1, the optimizer $\mathcal{S}_{\text{in}}^*$ is expected to
 149 be the more InD-like portion of the wild set, while
 150 $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ gives the corresponding candi-
 151 date OOD portion.

152 Directly solving (4) is computationally demanding
 153 for large wild sets, so Medix uses a greedy leave-
 154 one-out approximation summarized in Algorithm 1.
 155 The procedure iteratively removes samples whose
 156 gradients most improve alignment between the re-
 157 maining wild set and the InD reference.

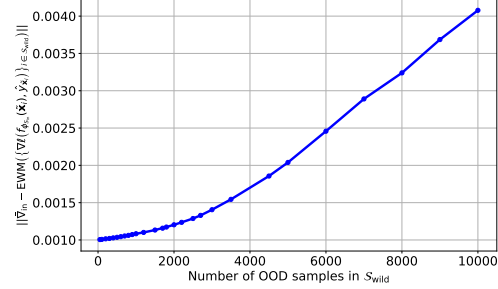


Figure 1: Median-gradient distance deviation as additional OOD samples are included in $\mathcal{S}_{\text{wild}}$.

158 At each iteration, Medix computes the EWM of
 159 the gradients for the current retained subset $\mathcal{S}$ and compares it with the InD average gradient $\bar{\nabla}_{\text{in}}$.
 160 We write d_t for this L_2 distance, $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. For each candidate sample $i \in \mathcal{S}$,
 161 the algorithm evaluates the leave-one-out distance $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ and the resulting drop
 162 $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. The k samples with the largest drops are treated as candidate
 163 OOD points and removed. Iteration continues until no substantial decrease remains or the maximum
 164 number of iterations is reached. The stopping criterion requires the change in L_2 distance to fall
 165 below the threshold ϵ . This rule reflects the pattern in Figure 1: once enough OOD samples have
 166 been removed, the retained set becomes close to the InD reference, and continuing to remove points
 167 risks discarding InD samples rather than improving the filter.

168 3.2 Training the OOD Detector with Candidate Outliers

169 Once Medix extracts the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$, we train the OOD detector g_θ to separate labeled
 170 InD samples from these candidates, following the protocol of Du et al. (2024a). The objective uses
 171 a zero threshold decision boundary, assigning positive outputs to labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and
 172 negative outputs to candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. The loss is:

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \quad (5)$$

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

173 Because the 0/1 loss is non-differentiable, implementation uses a differentiable sigmoid-based binary
 174 loss as a smooth surrogate for tractable optimization. The training objective also retains the InD risk
 175 from Equation 2, so detector learning does not discard the original classification task. In the evaluated
 176 settings, this combined objective supports generalization of g_θ to OOD samples drawn from $\mathbb{P}_{\text{out}}$.

177 4 Theoretical Analysis

178 We next state theoretical guarantees for the Medix filtering stage. The results bound misclassification
 179 rates for both InD and OOD points under the stated distributional assumptions, clarifying when
 180 EWM-based filtering can be expected to remain reliable for unlabeled mixtures. Detailed proofs
 181 appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

182 Theorem 4.1 decomposes the inlier error into two effects that are not tied to a particular dataset: a
 183 *concentration term*, which decreases with sample size and captures fluctuations of InD gradients,
 184 and a *contamination term*, which measures the influence of OOD samples on the EWM. The bound
 185 remains controlled when $\pi < 0.5$, reflecting the median’s robustness to moderate contamination. We
 186 then consider the complementary error: *how many true outliers are retained rather than flagged?*
 187 The next theorem bounds this OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

Theorem 4.2 complements the previous result by bounding the fraction of OOD points retained as InD. Together, Theorems 4.1 and 4.2 give a two-sided account of when Medix filtering is reliable for unlabeled mixtures. The inlier and outlier error rates are governed by the following effects:

Contamination Effect. In Theorem 4.1, the median-based criterion remains stable for mixed wild data when $\pi < 0.5$. The $\pi/[2(1 - \pi)]$ term grows with the OOD proportion and captures the extent to which OOD gradients can skew the EWM. Theorem 4.2 contains the corresponding penalty $\frac{1-\pi}{2\pi}$, which reflects the difficulty of isolating OOD points when they are less prevalent.

Concentration Effect. Both bounds use coordinate-wise sub-Gaussian gradient behavior. For InD data, concentration near $\bar{\nabla}_{\text{in}}$ stabilizes the median in finite samples. For OOD points, concentration around μ_{out} quantifies the probability that an OOD sample appears compatible with the retained set. These terms decay as $1/\sqrt{m}$, where m is the relevant sample count.

Separation Effect. This term appears in Theorem 4.2 and quantifies the gradient-space separation needed to reject OOD samples. The exponential factor $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ captures the same general trade-off: larger separation makes it less likely that OOD gradients are retained by the filter.

Theorems 4.1 and 4.2 jointly show that, with high probability, median filtering can separate InD and OOD samples in unlabeled mixtures under the stated assumptions. InD removal error is controlled by contamination and concentration effects, while OOD retention error is controlled by separation, concentration, and reverse contamination terms. These guarantees make explicit the conditions under which Medix controls both error types.

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The coordinate-wise sub-Gaussian assumption enables the concentration bounds used in the main theoretical statements. Empirically, Figure 4a supports this assumption in the evaluated setting: the InD gradient histogram is bell-shaped and concentrated around the mean, consistent with light-tailed behavior. Figure 4b further compares empirical gradient quantiles with Gaussian quantiles in a Q-Q plot (Wilk and Gnanadesikan, 1968); the alignment with the 45-degree reference line suggests that the observed gradient distribution is close to Gaussian. Because Gaussian variables are sub-Gaussian, this evidence supports the assumption for the studied models and data. *We also provide a looser version of these bounds, which removes the sub-Gaussian assumption and holds under merely bounded second moments. This version is presented in Theorem C.3 (Appendix C.3). While the rates degrade, the core robustness guarantee of Medix still holds in this broader setting.*

5 Experiments

This section evaluates Medix across multiple InD-OOD dataset pairs and compares it with 20 widely used baselines spanning different training assumptions. All experiments are run on hardware equipped with NVIDIA A100-SXM4-80GB GPUs. The code needed to reproduce the reported results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. We follow the experimental protocol of Katz-Samuels et al. (2022a), which studies OOD detector learning with wild data and provides a direct basis for comparison. CIFAR-10 and CIFAR-100 serve as the InD datasets ($\mathbb{P}_{\text{in}}$). For OOD evaluation, we use a suite of natural image datasets—PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015)—as OOD distributions ($\mathbb{P}_{\text{out}}$). To instantiate wild data ($\mathbb{P}_{\text{wild}}$) under this protocol, we mix a subset of InD data ($\mathbb{P}_{\text{in}}$) with the chosen OOD data ($\mathbb{P}_{\text{out}}$) using the default mixing parameter $\pi = 0.5$. For example, with PLACES365 as the OOD test set, the corresponding wild mixture combines CIFAR samples with PLACES365 samples and is tested against PLACES365 as OOD. The same construction is applied across all OOD datasets and baselines. Each CIFAR InD dataset is split into two halves: the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, and the remaining images contribute to the wild mixture $\mathcal{S}_{\text{wild}}$. For gradient computation, we use penultimate-layer weights, which prior work has found informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We report three standard metrics that capture both OOD rejection and retained InD performance: (1) False Positive Rate (FPR95 $\downarrow$) of OOD samples when the True Positive Rate of

234 InD samples is 95%, (2) Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and
 235 (3) InD Classification Accuracy (InD Acc $\uparrow$).

236 **Baselines.** The baselines cover two broad regimes: methods trained only on InD data and methods
 237 that also use wild or auxiliary data. For InD-only methods ($\mathbb{P}_{\text{in}}$), we compare against established OOD
 238 scores including Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang
 239 et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct
 240 (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic
 241 et al., 2022); these use models trained with softmax cross-entropy. We also include contrastive-loss
 242 methods such as CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022) to cover a wider class of
 243 detector backbones. For methods that use InD together with wild or auxiliary data, we compare
 244 with Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al.,
 245 2020b), which discourage confidence or encourage energy on outlier data. We also include WOODS
 246 (Katz-Samuels et al., 2022a), which introduced wild unlabeled data for OOD detection through
 247 constrained optimization, and recent baselines including CONJ (Peng et al., 2024) and DRL (Zhang
 248 et al., 2024).

249 5.2 Experimental Setup

250 Following WOODS (Katz-Samuels et al., 2022a), the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ is a Wide ResNet (Zagoruyko,
 251 2016) with 40 layers and width factor 2. We train it with stochastic gradient descent, momentum
 252 0.9, weight decay 0.0005, and initial learning rate 0.1 for 100 epochs, using cosine learning rate
 253 decay, batch size 128, and dropout 0.3. Medix hyperparameters ϵ and k are selected from $\{5\text{e-}5,$
 254 $5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$, respectively, accounting for dataset size and optimizing
 255 OOD performance. The OOD classifier g_{θ} is initialized from the pretrained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and
 256 augmented with a linear binary-classification layer on penultimate-layer features. We fine-tune for
 257 100 epochs with learning rate 0.001 using Equation (5). The binary classification loss is combined
 258 with the InD classification loss, with weight 10 on the binary component, and all other training
 259 parameters match those for $\phi_{\mathcal{S}_{\text{in}}}$.

Table 1: OOD detection performance for Medix and baselines with CIFAR-10 as InD data. Results are averaged over five runs, with the best values highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$	
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average			
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$		
	Using $\mathbb{P}_{in}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84	
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84	
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84	
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84	
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84	
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84	
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84	
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84	
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17	
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19	
	Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12	
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24	
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74	
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58	
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64	

260 5.3 Results

261 Table 2 reports the main CIFAR-100 results, where Medix achieves the strongest OOD detection
 262 performance among the compared methods. Two patterns are most relevant to the broader wild-data
 263 setting. First, methods trained with both InD and wild data generally outperform methods trained only
 264 on InD data, indicating that unlabeled mixtures can provide useful deployment-relevant information.
 265 Relative to KNN+, Medix reduces FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES.
 266 Second, Medix also improves over methods that already use wild data: on CIFAR-100, it reaches an
 267 average FPR95 of 5.42%, a 1.32% improvement over WOODS, while maintaining an InD accuracy
 268 of 73.33%. The small accuracy gap is expected because Medix is trained with 25,000 labeled InD
 269 samples, whereas baselines that do not use wild data use the full 50,000-image CIFAR-100 training
 270 set. Table 1 gives the corresponding CIFAR-10 results. In this setting, Medix again outperforms all
 271 listed baselines and improves over WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES in
 272 FPR95.

Table 2: OOD detection performance for Medix and baselines with CIFAR-100 as InD data. Results are averaged over five runs, with the best values highlighted in **bold**.

Methods	OOD Datasets										InD ACC $\uparrow$	
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average	
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29
Using $\mathbb{P}_{in}$ only												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{out}$												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10

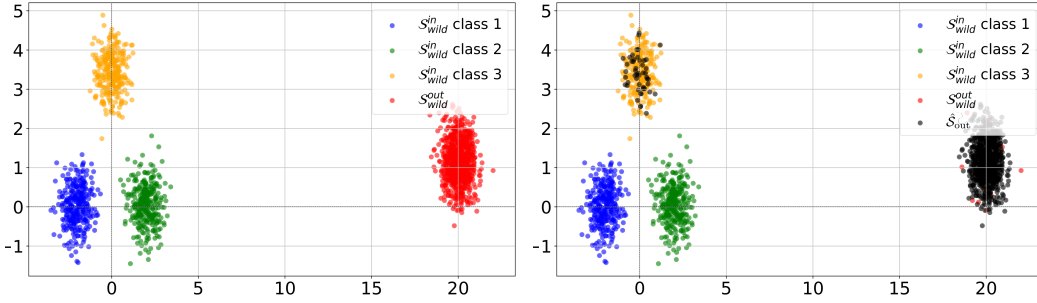


Figure 2: Illustrative application of Medix to unlabeled wild data. (a) InD samples S_{wild}^{in} and OOD samples S_{wild}^{out} in the wild mixture, with inliers drawn from three multivariate Gaussian distributions. (b) Outliers S_{out} selected by Medix (black); the selected set contains InD data S_{wild}^{in} at an error rate of only 12.5%.

A representative visual example of Medix. We also inspect Algorithm 1 as a standalone outlier-extraction procedure on a simple 2-dimensional synthetic mixture. The goal of this simulation is interpretability rather than added benchmark scope. The InD data are sampled from three multivariate Gaussian distributions corresponding to three classes, with means $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$. All three classes use covariance $0.25 \cdot I$, and the OOD data are drawn from $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows that the selected outliers largely align with the ground truth: Medix flags 87.5% of actual OOD samples as outliers in this illustrative setting.

Additional studies and insights. Appendix A provides additional experiments that probe the scope of the method under the same overall framework: (1) hyperparameter sensitivity (Appendix A.2), showing that Medix remains robust across the tested ϵ and k choices; (2) EWM versus geometric median (Appendix A.1), showing that EWM is more sensitive to the examined distributional shifts and therefore better suited to our filter; (3) comparisons with methods using competitive contrastive learning objectives (Appendix A.3); (4) evaluation in a large-scale unseen OOD setting with $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4); (5) computation and memory profiling for Medix filtering (Appendix A.6); (6) analysis of pseudo-label quality (Appendix A.5); and (7) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has become an active research area spanning many assumptions about model access, training data, and test-time shifts (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). A large family of methods detects OOD data by assigning scores to inputs, including distance-based methods (Lee

et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

A second line of work modifies training through regularization (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). These methods may reduce model confidence (Lee et al., 2017; Hendrycks et al., 2019) or increase energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data. Many such approaches assume that an auxiliary OOD dataset is available and sufficiently clean for training.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the labeled-data-only assumption by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), although they do not provide an explicit outlier-identification mechanism. Du et al. (2024a,b) do introduce filtering, but their thresholding strategy differs from the median-centric criterion studied here. Moreover, Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, where each batch has a fixed InD/OOD ratio. In large outsourced or deployment-collected datasets, this structure may not be available because samples are mixed at the dataset level. Medix is designed for this broader dataset-level mixture setting. Positive-unlabeled learning also studies classifiers trained with positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). In contrast, our setting requires both identifying candidate outliers from unlabeled data and training a dedicated OOD detector. Medix addresses this by using a median-centric filter with provably low error under the stated assumptions.

7 Conclusions

We introduced Medix, a median-centric framework for OOD detection with unlabeled in-the-wild data. By using the robustness of the median in gradient space, Medix filters candidate outliers from mixed unlabeled data and uses them to train an OOD detector. The theoretical analysis bounds inlier misclassification under OOD contamination up to 50%, with error controlled by concentration and contamination terms, and also bounds OOD retention under a separation condition. Across the evaluated benchmarks, Medix improves over 20 baselines, reducing average FPR95 by 40.98% relative to strong baselines such as KNN+ and outperforming methods such as WOODS and DRL, while achieving an outlier extraction error rate as low as 12.5%. These results support median-based filtering as a broadly applicable tool for wild-data OOD detection within the assumptions and experimental conditions studied here.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

A natural design question is whether Medix should use the element-wise median or the geometric median (Acharya et al., 2024) for gradient-based outlier filtering. We examine this choice in a preliminary CIFAR-10/SVHN experiment, with CIFAR-10 as InD and SVHN as OOD. The wild set $\mathcal{S}_{\text{wild}}$ begins with 5k CIFAR-10 samples disjoint from the training data used to learn $\phi_{\mathcal{S}_{\text{in}}}$ and compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN OOD samples and track both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the wild-data median gradient and the proportion of OOD samples removed. Figure 3 shows that, for the same number of OOD samples in the wild set, the element-wise median identifies a higher proportion of OOD samples than the geometric median. In this setting, EWM is more sensitive to the relevant distributional shift, making it the more effective aggregation choice for Medix filtering.

Table 3: Sensitivity of OOD detection performance to Medix hyperparameters ϵ and k .

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We assess sensitivity to the Medix hyperparameters ϵ and k using CIFAR-100 as InD and SVHN as OOD. Table 3 shows that Medix is stable across the tested values. With $\epsilon = 0.005$ and $k = 20000$, Medix achieves an FPR95 of 0.16. Reducing ϵ to 0.0005 or halving k to 10000 yields FPR95 values of 0.20 and 0.68, respectively, indicating gradual degradation rather than a sharp failure. Thus, although hyperparameter choice matters, Medix remains effective over a range of settings and continues to outperform the listed baselines. This reduced dependence on exact tuning is valuable in deployment-like settings where exhaustive validation may be limited.

A.3 Comprehensive comparison with recent competitive methods

We extend the comparison to baselines with diverse contrastive or representation-learning objectives, including SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). Although Medix does not use these contrastive objectives, including them broadens the comparison beyond standard scoring and regularization methods. We also include recent methods CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). For this experiment, CIFAR-100 is the InD dataset and the backbone is ResNet-34¹, following Ming et al. (2023); Zhang et al. (2024). The model is trained with stochastic gradient descent, momentum 0.9, and weight decay 10^{-4} for 500 epochs. The initial learning rate is 0.5 with cosine scheduling and the batch size is 512. We use the checkpoints provided by Ming et al.

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

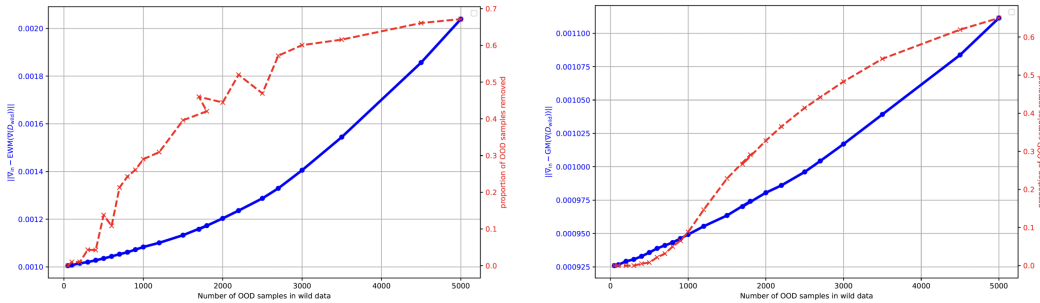


Figure 3: Comparison of element-wise median (EWM) and geometric median (GM) as robust aggregation choices for filtering.

Table 4: OOD detection performance for Medix and recent competitive baselines with CIFAR-100 as InD data. Results are averaged over five runs, with best values highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

(2023)². As shown in Table 4, Medix outperforms these competitive baselines, including those trained with contrastive learning objectives, with average FPR95 8.68% and average AUROC 97.76%. This comparison indicates that the median-filtering mechanism remains competitive even against stronger representation-learning alternatives, while also providing the theoretical error control developed above.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next evaluate Medix in a larger-scale unseen OOD setting. This setting broadens the experimental stress test in two ways: (1) the wild OOD component ($\mathbb{P}_{\text{out}}$) is large, increasing data complexity and computational demand, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is intentionally different from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$. We use CIFAR-100 as labeled InD data, 300K Random Images Hendrycks et al. (2019) as unlabeled wild OOD data, and SVHN as the OOD test data. The setting is challenging because the large-scale wild OOD data (300K Random Images) is substantially different from the test OOD data (SVHN). We compare against baselines that also use wild data during training. Table 5 shows that Medix achieves lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.6) than the baselines, indicating that the learned detector remains useful when the training-time and test-time OOD sources differ.

Table 5: OOD detection performance in the large-scale unseen OOD setting.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

We also examine whether pseudo-label quality, especially softmax confidence in $\hat{y}_{\tilde{x}_i}$, materially affects OOD filtering, and whether removing low-confidence pseudo-labels improves robustness. The experiment uses CIFAR-10 as InD data and LSUN-Resize as OOD data. We compute softmax probabilities for each pseudo-labeled sample and remove samples below confidence threshold 0.6. This pre-filtering step removes 15.98% of samples from the training set.

The pre-filtered and unfiltered variants perform almost identically. Both obtain FPR95 of 0.01%, while AUROC changes slightly from 99.98% to 99.99% with filtering. Thus, in this setting, removing low-confidence pseudo-labels does not materially change OOD detection performance, suggesting that Medix is resilient to this level of pseudo-label noise and that additional confidence filtering is unlikely to be beneficial.

²<https://github.com>

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We next profile whether the Medix filtering stage is computationally feasible and memory efficient for larger unlabeled sets. Profiling is performed on an NVIDIA A100-SXM4-80GB GPU with approximately 15,000 unlabeled samples for each InD–OOD pair. The InD datasets are CIFAR-10 and CIFAR-100, and the OOD dataset is LSUN-Resize. Table 6 shows modest GPU memory use and tractable wall-clock time for thousands of unlabeled samples. These measurements indicate that the filtering stage can be run at the evaluated scale without excessive memory overhead.

Table 6: Runtime and memory profile for Medix filtering on an NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix also differs from semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in its problem formulation and use of unlabeled data. For instance, OpenML Saito et al. (2021) addresses outliers in semi-supervised learning (SSL) while training a standard classifier. Our setting instead focuses on detecting OOD samples from unlabeled wild data and then training a dedicated OOD detector. Thus, SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to preserve classification performance in the presence of outliers, whereas Medix targets OOD detection when labeled OOD data is unavailable.

Second, SSL open-set methods typically treat OOD samples as noise to be suppressed during classifier training. Medix instead treats candidate OOD samples as useful training signal for a separate detector after filtering them from the wild mixture.

Finally, the techniques differ. OpenMatch relies on consistency regularization across stochastic transformations, implemented through soft constraints on classifier outputs to encourage smooth decision boundaries. Medix instead uses median-based filtering, leveraging the robustness of the median to identify samples that shift the wild-gradient aggregate away from the InD reference. The theory analyzes contamination and concentration effects to bound filtering error in this OOD detection setting.

B Broader Impacts and Limitations

As machine learning systems are deployed in less curated environments, OOD detection remains an important component of reliability and safety. Medix contributes a median-centric framework for using unlabeled wild data in this setting, and the same general formulation is relevant to application areas where unexpected inputs can have practical consequences, such as healthcare, autonomous systems, and finance. By improving detector training without requiring a clean auxiliary OOD set, the approach addresses one deployment barrier for robust machine learning. Future work could combine Medix with generative models for outlier synthesis or adapt the filtering procedure to dynamic environments where OOD distributions evolve over time.

A limitation of Medix is that it assumes access to an unlabeled mixture containing both InD and OOD samples; in practice, such data may be noisy, corrupted, or unevenly sampled. This assumption is common in robust learning formulations of deployment uncertainty, but it still limits the settings in which the guarantees apply directly. We make these conditions explicit in the main theorems and discuss the geometric separation required for reliable filtering in Remark 4.3. We also extend the analysis in Theorem C.3 by removing the sub-Gaussian tail assumption, showing that a weaker guarantee remains available under broader moment conditions. Thus, the assumptions are simplifying but interpretable, and they indicate clear directions for extending Medix to noisier and more dynamic OOD environments.

704 C Proof of Theorem

705 For clarity across the theory, experiments, and appendix, Table 7 summarizes the main notation
 706 used in the paper. The table covers gradient quantities, sample sizes, filtering terms, and parameters
 707 appearing in the concentration bounds. Unless otherwise stated, bold symbols denote vectors, $\mathcal{X}$
 708 denotes the input space, and gradients are computed with respect to a fixed pretrained model f_ϕ .

Table 7: Notation summary for the wild-data OOD detection setting. Bold symbols denote vectors and $\mathcal{X}$ denotes the input space.

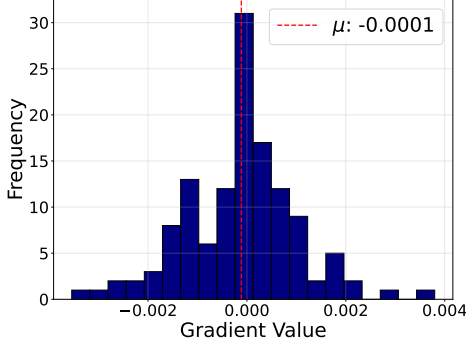
Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

709 C.1 Bound on the Inlier Misclassification Rate

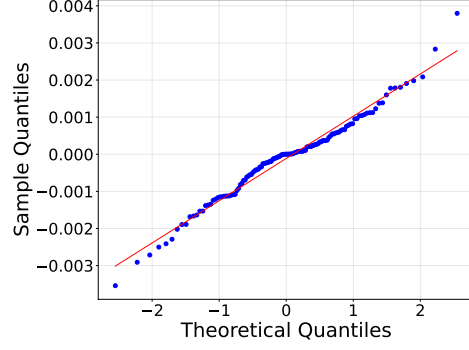
710 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 711 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 712 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 713 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

714 *Proof.* The proof has two steps that mirror the general filtering mechanism. In **Step 1**, we obtain
 715 a high-probability concentration bound for InD gradients around $\bar{\mathbf{v}}_{\text{in}}$ and call an InD point a *good*
 716 *inlier* when its gradient lies in this region. In **Step 2**, we use a “swapping” argument for the EWM
 717 objective: if the optimizer removed more than a $(2\eta + \frac{\pi}{2}(1 - \pi))$ fraction of *good inliers*, where η is
 718 the fraction of InD points outside the concentration region, then replacing bad or OOD points with
 719 good inliers would improve the objective and contradict optimality. Combining these steps yields the
 720 bound on ERR_{in} .



(a) Histogram of InD gradient values used to assess the light-tail assumption.



(b) Q-Q plot comparing InD gradient quantiles with Gaussian quantiles.

Figure 4: Empirical evidence that InD sample gradients in the evaluated CIFAR-100 setting are compatible with coordinate-wise sub-Gaussian behavior. (left) Histogram of gradient values showing concentration around the mean with light tails. (right) Q-Q plot comparing empirical InD gradient quantiles with Gaussian quantiles, supporting the sub-Gaussian modeling assumption.

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ denote the loss gradient, with respect to ϕ , for an InD sample $\mathbf{x}$. The theorem assumes that each coordinate is sub-Gaussian with variance proxy σ^2 and common mean vector $\bar{\nabla}_{\text{in}}$. Thus, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

The following norm relation connects coordinate-wise concentration to the ℓ_2 deviation:

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

Applying a union bound over the d coordinates in (6) gives

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ be the number of InD points in $\mathcal{S}_{\text{wild}}$. We call an InD point $\mathbf{x}$ *bad* when

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

The goal of this step is to show that only a small fraction of the m_{in} inliers are bad with high probability.

Using the sub-Gaussian assumption and the coordinate-wise union bound, define $p \in [0, 1]$ by

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here, $\mathbf{x}$ denotes a generic InD sample, and the m_{in} inliers are assumed i.i.d.

Let X count bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

735 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

736 Since X is a sum of bounded Bernoulli variables, Hoeffding's inequality implies that for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

737 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

738 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

739 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

740 Dividing by m_{in} converts the count bound into a bound on the bad-inlier fraction:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

741 Denote this fraction by η . With $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, with probability at least $1 - \delta$, no more
 742 than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients outside the radius $\epsilon\sqrt{d}$
 743 around $\bar{\nabla}_{\text{in}}$. These points are the *bad inliers*; all remaining inliers are *good inliers*.

744 **Step 2: Contradiction via a Swapping Argument** We next show that $\mathcal{S}^*$ cannot exclude too many
 745 InD points. Suppose, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

746 Then one can construct a set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than the EWM of $\mathcal{S}^*$, contradict-
 747 ing optimality.

748 Let m_ε count indices i for which x_i is either OOD or an InD point with gradient outside the closed
 749 ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

750 Let m_{in}^* denote the number of InD points retained by the candidate minimizer $\mathcal{S}^*$.

751 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

752 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

753 This value is the number of good inliers, i.e., points inside B_ε , that are excluded from $\mathcal{S}^*$ and can be
 754 swapped in.

755 Construct $\hat{\mathcal{S}}$ by removing m'_ε points in $\mathcal{S}^*$ outside B_ε , whether OOD or bad inliers, and inserting m'_ε
 756 good inliers from B_ε . This preserves cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

757 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

758 which is the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging gives:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

759 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

760 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

761 At most a η fraction of inliers lie outside B_ε , and the OOD fraction among all samples is π . Therefore
 762 the number of samples outside B_ε is bounded by:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

763 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

764 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

765 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

766 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

767 so points inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

768 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of
 769 gradients in $\mathcal{S}^*$, and let $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ be the corresponding values in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

770 Each inserted gradient lies in $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, while each removed gradient lies outside I_j .
 771 Thus, for some coordinate j^* , the swaps increase the number of values inside I_{j^*} from below $m/2$ to
 772 above $m/2$. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

773 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

774 For each remaining coordinate $j \neq j^*$, the swaps either preserve or increase the majority inside the
 775 corresponding interval, so the median distance from μ_j cannot increase.

776 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

777 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

778 This contradicts the assumption that $\mathcal{S}^*$ minimizes the objective and completes the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

779 □

780 C.2 Bound on the Outlier Retention Rate

781 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
 782 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
 783 (Appendix C.1), but instead of relying on inlier concentration, it exploits a separation assumption
 784 between the OOD and InD gradient means.

785 **Theorem C.2** (Outlier Retention Bound under Vector Separation). *Assume that for every OOD point*
 786 *$x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:*

787 **(I) Sub-Gaussianity.** *Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .*

788 **(II) Vector Mean Separation.** *The mean OOD gradient vector μ_{out} satisfies*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta \sqrt{d}, \quad \text{for some } \Delta > 0.$$

789 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

790 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

791 *Proof. Step 1: Concentration of OOD Gradients.*

792 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
793 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

794 has probability at most $p_{\text{out}}(\varepsilon)$.

795 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

796 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

797 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

798 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

799 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

800 which holds with probability at least $1 - \delta$.

801 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

802 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_{\varepsilon}^* > \frac{1 - \pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

803 Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_{\varepsilon} := \left\{g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}. \quad (46)$$

804 By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of
805 entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x))_j - \bar{\nabla}_{\text{in},j})^2] = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x))_j - \bar{\nabla}_{\text{in},j})^4] \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1 - \pi)}}_{\text{Contamination term}}.$$

Differences with the original bound. The sub-Gaussian bound uses the decay term

$$2 \exp\left(-\frac{\varepsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

which decreases exponentially with dimension d and is therefore tight in high-dimensional regimes under the stronger tail assumption. The finite-fourth-moment bound replaces this with

$$\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2}$$

which decays as $O(1/d)$. This rate is looser but applies under the broader condition of finite fourth moments, allowing heavier-tailed gradient coordinates. Both bounds retain the same finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

which shrinks with m_{in} and is independent of the tail assumption.

830 *Proof.* The proof follows the same two-step structure as the sub-Gaussian result, but with a broader
 831 concentration argument: (1) bound the fraction of “bad” InD samples, those far from ∇_{in} , without
 832 sub-Gaussian tails; and (2) use the same swapping argument to control the inlier misclassification
 833 rate. The key change is replacing the exponential sub-Gaussian tail bound with a moment-based
 834 bound derived from the finite fourth moment assumption.

835 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

836 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 837 and we assume ∇_{in} is the population mean. We call an InD sample “bad” when:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

838 and “good” otherwise. The aim is to bound the fraction of bad inliers among the m_{in} InD samples in
 839 S_{wild} .

840 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, the ℓ_2 norm can be controlled through squared coordinate deviations.
 841 Assume the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$ and satisfy:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

842 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

843 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

844 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

845 Using Chebyshev’s inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

846 We bound $\mathbb{P}(\|v\|_2 > \epsilon \sqrt{d})$ by considering the equivalent squared-norm event $\|v\|_2^2 > \epsilon^2 d$:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

847 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

848 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}\left(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

849 Let X count bad inliers among the m_{in} InD samples in S_{wild} . Each InD sample is bad with probability
850 at most p , so:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

851 Because $X = \sum_{i=1}^{m_{\text{in}}} I_i$, with $I_i = 1$ for a bad InD sample and $I_i = 0$ otherwise, Hoeffding's
852 inequality for bounded Bernoulli variables gives:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

853 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

854 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

855 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

856 This η is an upper bound on the fraction of InD samples whose gradients deviate from ∇_{in} by more
857 than $\epsilon\sqrt{d}$ under only finite fourth moments and coordinate independence.

858 **Step 2: Contradiction via a Swapping Argument** We next show that $\mathcal{S}^*$ cannot exclude too many
859 InD points. Suppose, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

860 Then one can construct a set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\mathbf{V}}_{\text{in}}$ than the EWM of $\mathcal{S}^*$, contradict-
 861 ing optimality.

862 Let m_ε count indices i for which x_i is either OOD or an InD point with gradient outside the closed
 863 ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\mathbf{V}}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (69)$$

864 Let m_{in}^* denote the number of InD points retained by the candidate minimizer $\mathcal{S}^*$.

865 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

866 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

867 This value is the number of good inliers, i.e., points inside B_ε , that are excluded from $\mathcal{S}^*$ and can be
 868 swapped in.

869 Construct $\hat{\mathcal{S}}$ by removing m'_ε points in $\mathcal{S}^*$ outside B_ε , whether OOD or bad inliers, and inserting m'_ε
 870 good inliers from B_ε . This preserves cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

871 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

872 which is the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging gives:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

873 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

874 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

875 At most a η fraction of inliers lie outside B_ε , and the OOD fraction among all samples is π . Therefore
 876 the number of samples outside B_ε is bounded by:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

877 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

878 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

879 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

880 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

881 so points inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

882 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of
883 gradients in $\mathcal{S}^*$, and let $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ be the corresponding values in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

884 Each inserted gradient lies in $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, while each removed gradient lies outside I_j .
885 Thus, for some coordinate j^* , the swaps increase the number of values inside I_{j^*} from below $m/2$ to
886 above $m/2$. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

887 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

888 For each remaining coordinate $j \neq j^*$, the swaps either preserve or increase the majority inside the
889 corresponding interval, so the median distance from μ_j cannot increase.

890 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

891 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

892 This contradicts the assumption that $\mathcal{S}^*$ minimizes the objective and completes the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

893 □